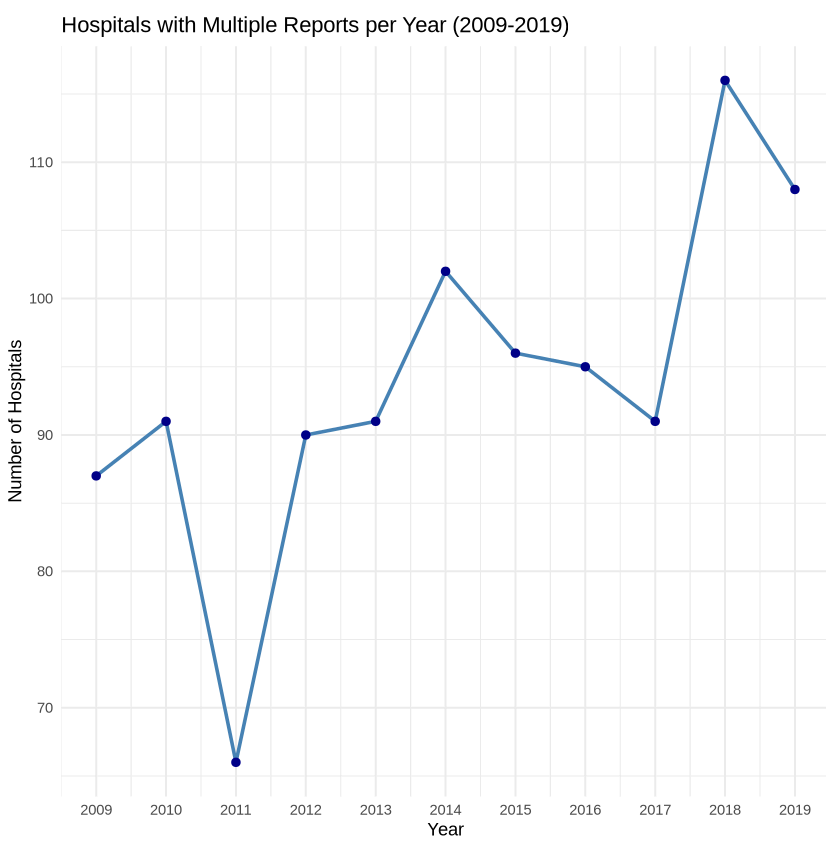
```
Loading required package: pacman
```

Question 1: From 2009-2019, how many hospitals filed more than one report in the same year?

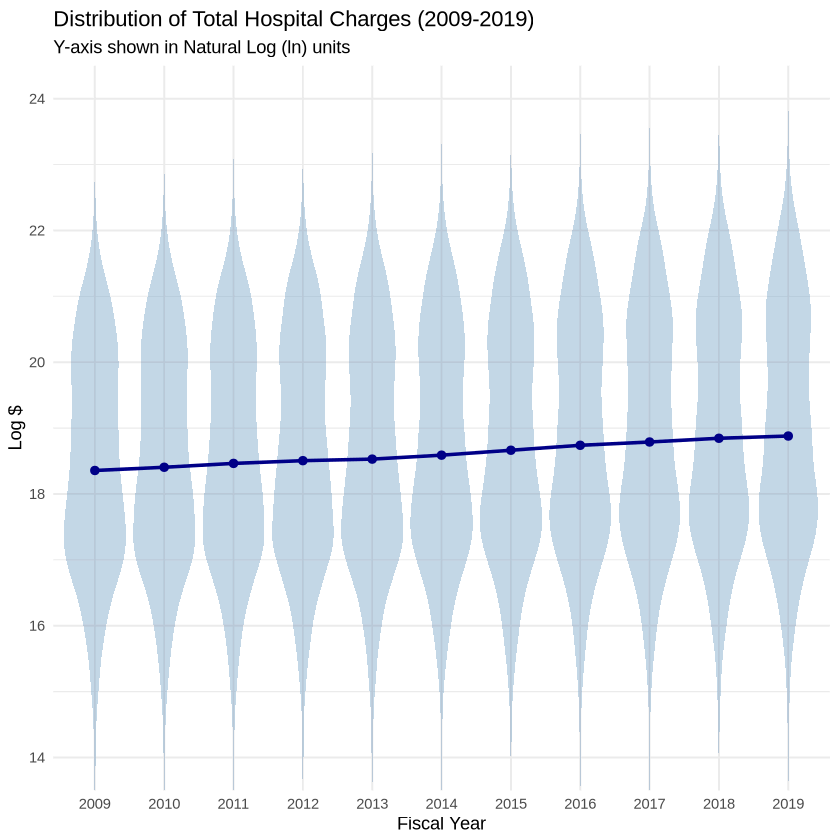
```
Rows: 78508 Columns: 47
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (12): provider_number, fy_start, fy_end, date_processed, date_created, d...
dbl (34): report, status, year, beds, tot_charges, net_pat_rev, tot_discount...
lgl  (1): npi

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 85496 Columns: 48
── Column specification ────────────────────────────────────────────────────────
Delimiter: "\t"
chr (12): provider_number, fy_start, fy_end, date_processed, date_created, d...
dbl (35): report, status, year, beds, tot_charges, tot_discounts, net_pat_re...
lgl  (1): npi

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```



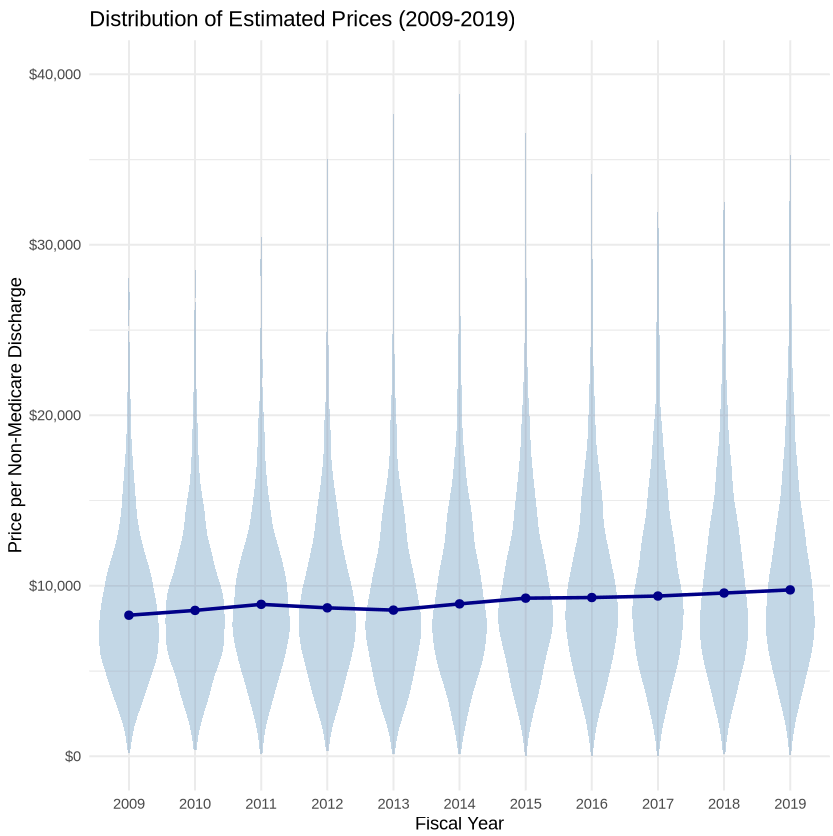
Question 2: After removing/combining multiple reports, how many unique hospital IDs (Medicare provider numbers) exist in the data?

```
Total unique Medicare provider numbers in cleaned data: 6897
```

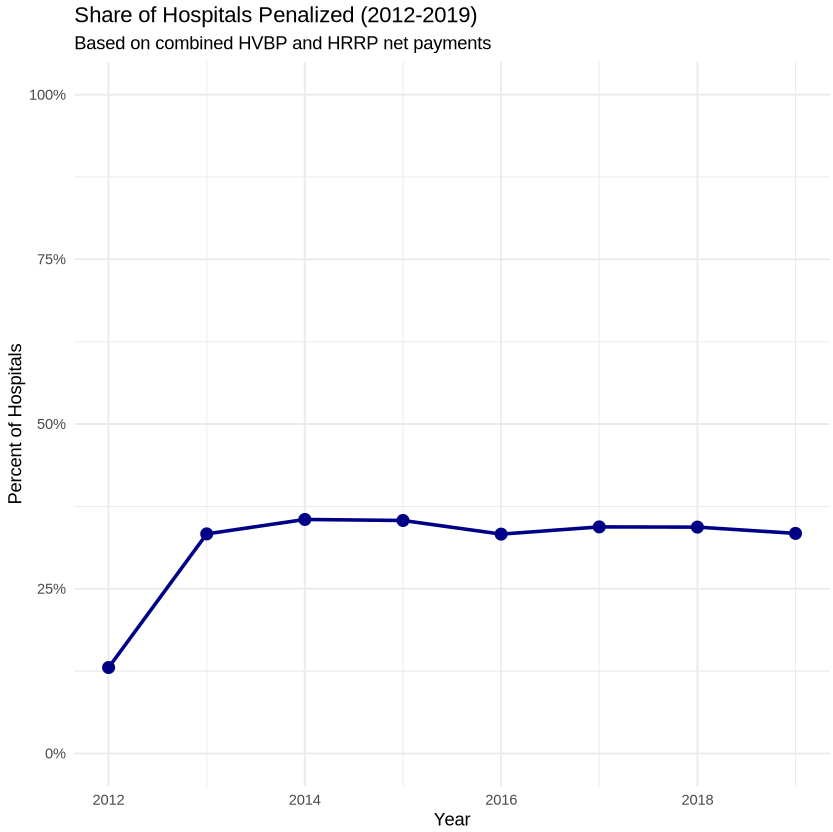
Question 3: From 2009-2019, what is the distribution of total charges (tot_charges in the data) in each year?



Question 4: From 2009-2019, what is the distribution of estimated prices in each year?



Question 5: What share of hospitals are penalized under the HRRP/VBP? (2012-2019)



Question 6: Provide a summary of OLS estimates of the effect of net penalties on price changes.

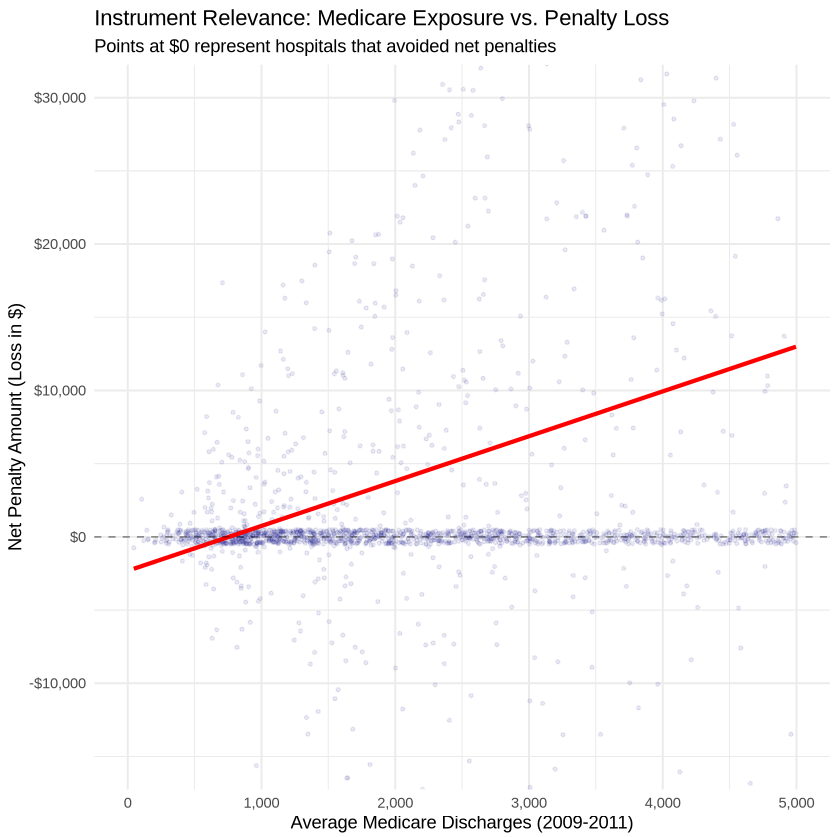
```
NOTE: 5 observations removed because of NA values (RHS: 5).

NOTE: 5 observations removed because of NA values (RHS: 5).

NOTE: 16 observations removed because of NA values (RHS: 16).
```

A etable_df: 11 × 4

		m1_ols	m2_ols	m3_ols
	<chr>	<chr>	<chr>	<chr>
1	Dependent Var.:	price_change	price_change	price_change
2				
3	Constant	84.98 (75.02)	66.95 (75.86)	-65.01 (93.50)
4	penalty_amt	-0.0045* (0.0019)	-0.0044* (0.0019)	-0.0039* (0.0019)
5	avg_beds		0.0751 (0.0476)	0.0668 (0.0478)
6	avg_mcaid_disch			0.0842* (0.0350)
7	_____	_____	_____	_____
8	S.E. type	IID	IID	IID
9	Observations	2,508	2,508	2,497
10	R2	0.00228	0.00327	0.00559
11	Adj. R2	0.00188	0.00247	0.00439

Question 7: Provide a scatterplot of net penalty against pre-2012 Medicare discharges.

```
`geom_smooth()` using formula = 'y ~ x'
Warning message:
“Removed 611 rows containing non-finite values (`stat_smooth()`).”
Warning message:
“Removed 611 rows containing missing values (`geom_point()`).”
```



Question 8: Provide a summary of the first stage and reduced-form results using pre-penalty Medicare discharges as an instrument for net penalties.

```
NOTE: 5 observations removed because of NA values (LHS: 5).

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NOTE: 17 observations removed because of NA values (LHS: 5, RHS: 12).

NOTE: 12 observations removed because of NA values (RHS: 12).
```

A etable_df: 12 × 4

		fs_1	fs_2	fs_3
	<chr>	<chr>	<chr>	<chr>
1		Baseline	+ Beds	+ Medicaid
2	Dependent Var.:	penalty_amt	penalty_amt	penalty_amt
3				
4	Constant	-1,253.9*** (366.3)	-1,255.2*** (366.5)	-1,240.6*** (372.8)
5	avg_mcare_disch	0.6567*** (0.0766)	0.6557*** (0.0771)	0.6720*** (0.0887)
6	avg_beds		0.0197 (0.1576)	0.0213 (0.1581)
7	avg_mcaid_disch			-0.0462 (0.1307)
8	_____	_____	_____	_____
9	S.E. type	IID	IID	IID
10	Observations	2,523	2,523	2,511
11	R2	0.02832	0.02833	0.02836
12	Adj. R2	0.02793	0.02755	0.02720

A etable_df: 12 × 4

		rf_1	rf_2	rf_3
	<chr>	<chr>	<chr>	<chr>
1		Baseline	+ Beds	+ Medicaid
2	Dependent Var.:	price_change	price_change	price_change
3				
4	Constant	-1,123.2 (773.5)	-1,128.4 (773.9)	-326.7 (568.7)
5	avg_mcare_disch	0.3678* (0.1618)	0.3637* (0.1627)	0.2629. (0.1354)
6	avg_beds		0.0798 (0.3332)	0.0743 (0.2414)
7	avg_mcaid_disch			-0.0540 (0.1995)
8	_____	_____	_____	_____
9	S.E. type	IID	IID	IID
10	Observations	2,528	2,528	2,516
11	R2	0.00204	0.00206	0.00185
12	Adj. R2	0.00165	0.00127	0.00066

Question 9: Provide a summary of IV estimates of the effect of net penalties on price changes.

```
NOTE: 5 observations removed because of NA values (IV: 5/0).

NOTE: 5 observations removed because of NA values (IV: 5/0).

NOTE: 16 observations removed because of NA values (RHS: 11, IV: 5/0).
```

A etable_df: 11 × 4

		m1_iv	m2_iv	m3_iv
	<chr>	<chr>	<chr>	<chr>
1		Baseline	+ Beds	+ Medicaid
2	Dependent Var.:	price_change	price_change	price_change
3				
4	Constant	119.5 (89.62)	108.5 (89.89)	10.26 (99.90)
5	penalty_amt	-0.0008 (0.0056)	0.0002 (0.0056)	0.0088 (0.0058)
6	avg_beds		0.0790. (0.0479)	0.0752 (0.0483)
7	avg_mcaid_disch			0.1095** (0.0370)
8	_____	_____	_____	_____
9	S.E. type	IID	IID	IID
10	F-test	0.01951	1.3719	4.0069
11	Observations	2,508	2,508	2,497

Question 10: Briefly explain the "Local" ATE in the context of your estimates. How might a local effect differ from an overall ATE in this setting?

The IV estimate represents a Local ATE because it identifies the causal impact of net penalties on price changes specifically for the hospitals whose penalty status was directly influenced by their volume of Medicare discharges (the instrument). UPDATE THIS W NEW VALUES The local effect (0.3895) is significantly larger and more statistically meaningful than the overall ATE estimated by OLS (0.0058).